JalTaap

A heat, rain and water safety helper for India's outdoor workers

Is it solving a real problem? What can it do? Why should it win? Answered in simple words.

414 days in 2026 (counting each city separately, across 63 Indian cities) that were dangerous because of humid heat, but never met a 40 °C heatwave rule.	**0 of 42** dangerous heat days in Kolkata in 2026 that a 40 °C rule would have caught.
46 °C reached in several parts of Delhi on 22 May 2026.	**5.8%** of working hours India is projected to lose to heat by 2030, equal to 34 million jobs.

Sources: our own backtest on Open-Meteo modelled weather data (1 April to 31 July 2026, daytime hours); UNI news (Delhi 46 °C); ILO via The Quint (working hours).

In one line: other systems warn a whole city about the weather. JalTaap tells one worker, in their own language and by voice, exactly when it is safe to work today, and makes sure their complaint reaches the office that can fix it.

1. Is this a real-world problem?

Yes. It is one of the biggest everyday safety problems in Indian cities, and it gets worse every year.

Who suffers

People who work outside all day: construction workers, delivery riders, street vendors, loaders, traffic police, sanitation workers. Many earn daily wages, use a basic Android phone, read slowly or not at all, and cannot stop working just because it is hot.

What goes wrong today

- **Heat warnings look only at temperature.** India usually calls a heatwave at 40 °C or more. But humid heat can kill at lower temperatures: 36 °C with 65% humidity is already near the limit a healthy body can cool itself (Penn State lab research). These humid days are mostly missed.
- **Warnings are for the whole city, not for you.** "Heatwave in Delhi" does not tell a mason in Najafgarh when it is safe to lift heavy loads today, or how much water to drink.
- **Warnings do not reach workers.** They sit in English apps, news and government websites. Workers use WhatsApp and speak Hindi.
- **Monsoon flooding catches people by surprise.** Delhi Traffic Police listed 169 waterlogging spots in 2026. In 2024, three students died in a flooded basement in Old Rajinder Nagar, Delhi.
- **Complaints get lost.** Lists of flood spots change every year and some spots have no clear owner (71 Delhi spots had ownership disputes in 2025), so a report often reaches nobody who can act.
- **Water is a daily struggle too.** Delhi Jal Board estimates a shortfall of about 250 million gallons a day; families wait for tankers that may not come.

Proof with our own numbers

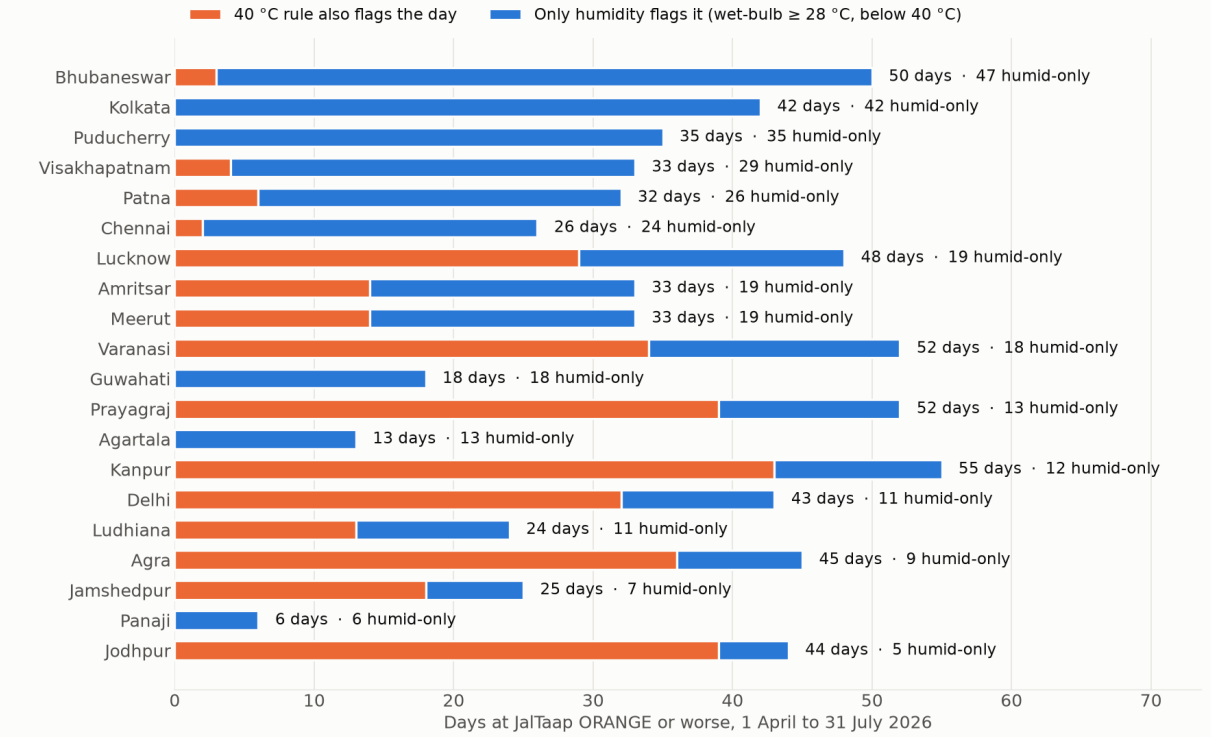
We tested JalTaap's humid-heat rule against real (modelled) weather for April to July 2026:

What we measured	Result
Delhi (Safdarjung): days dangerous by humidity that a 40 °C rule missed	11 days
63 Indian cities: city-days dangerous by humidity alone	414 city-days
Bhubaneswar: dangerous days, and how many a 40 °C rule caught	50 days, rule caught only 3
Kolkata: dangerous days, and how many a 40 °C rule caught	42 days, rule caught none
Day-ahead check (Delhi): ORANGE-or-worse days flagged one day early	35 of 43

All weather values are modelled data (Open-Meteo), not weather-station readings. We did not tune our thresholds to match news headlines.

Across 63 Indian cities, 414 city-days in 2026 were dangerous by humidity alone

A 40 °C heatwave rule never flags the blue days. Bhubaneswar had 50 dangerous days and the 40 °C rule caught only 3 of them.



Modelled data: Open-Meteo Historical Weather API (reanalysis). One point per city, 06:00-19:59 IST. Top 20 of 63 cities.

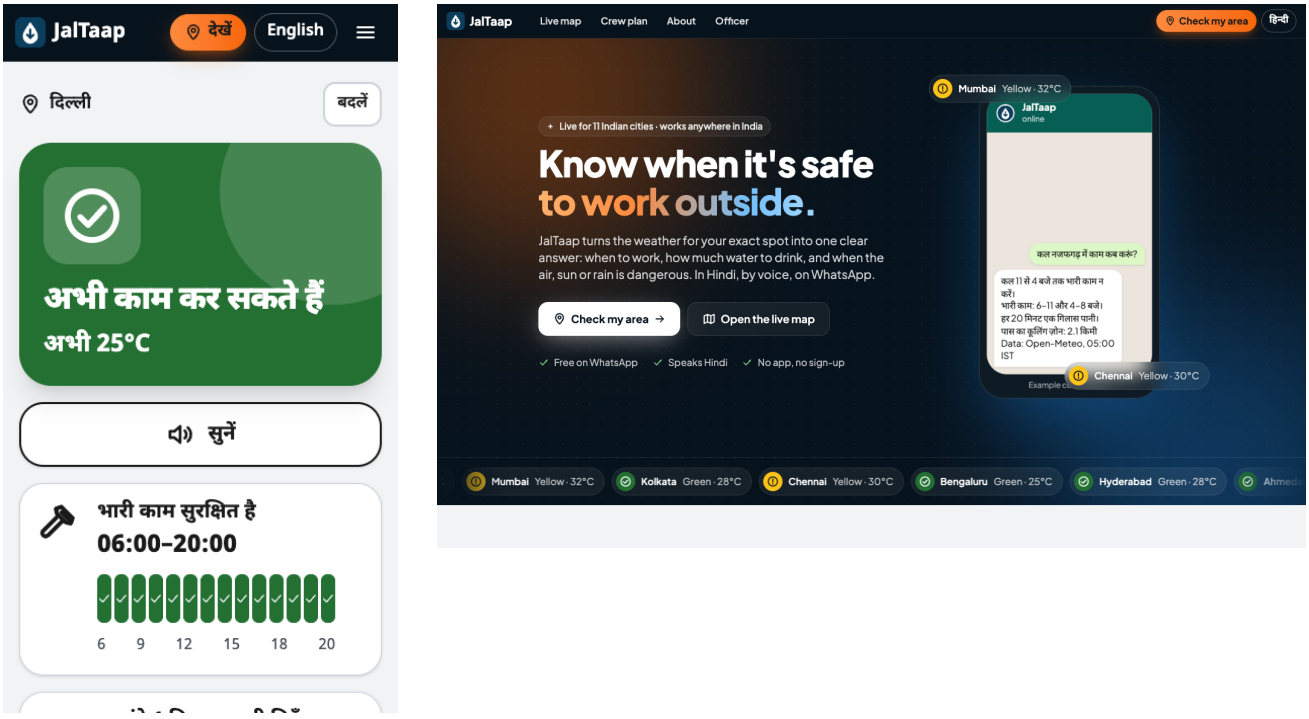
2. How JalTaap solves it

Four simple steps. The AI only talks; a tested calculator decides the risk, so the answer never depends on the AI guessing.

Step	What happens
1. Ask	The worker sends a message on WhatsApp, Telegram or the website, in Hindi, Hinglish or English. They can type, share their location pin, or send a voice note.
2. Calculate	JalTaap gets the forecast for that exact spot and works out the risk for every hour: heat plus humidity, hot nights, heat streaks, rain, air pollution and strong sun.
3. Answer	A short reply in their language: when heavy work is safe, when to stop, how much water to drink, the nearest cool place. It can be read aloud in a female Hindi voice. If someone shows heatstroke signs, the first line says "Call 112 now".
4. Act	Photos of flooding, leaks or tanker queues are checked by AI, put on the map, and sent to the office that owns that spot. Officers see their own queue and an action list.

What a worker sees

The worker screen shows one big colour, one big picture and a few words, plus a Listen button and a big Call 112 button. Here it is in Hindi:



Left: worker screen in Hindi ("You can work now", 25 °C, Listen, heavy-work hours). Right: the public website with live heat for Indian cities.

3. Everything JalTaap can do

For workers

Feature	What it does
Hour-by-hour safe-work plan	When heavy work is safe today, when to pause (12 to 3 on hot days), for any place in India.
Humid-heat warning	Uses wet-bulb temperature (heat plus humidity), so it catches dangerous humid days a 40 °C rule misses.
Personal risk	Heavy work, age 60+, pregnancy, illness or a hot night make the warning stronger for that person.
Water advice	How many glasses of water to drink every hour.
Nearest cool place	Cooling centres, water points and hospitals with distance and a directions link; never a place that is only planned.
Emergency first	Heatstroke signs put "Call 112 now" on line one, then simple first-aid steps.
Flood check and route check	Waterlogging risk for the next 3 hours near you or on your way, with the office that owns the spot.
Air quality and sun	An India AQI estimate and the UV index, with mask and head-cover advice when needed.
Alerts and morning brief	Subscribe once; get warned when your area turns dangerous (at most once in 6 hours) and a 5:30 am daily brief.
Report a problem	Send a photo of flooding or a leak, or just type "tanker did not come".
Privacy and STOP	Only language, a rough location and chosen alerts are kept. Sending STOP deletes everything.

For people who cannot read

Feature	What it does
Picture-first screen	One giant colour, a big icon and a few words. Colour is never used alone.
Listen button	Reads the plan aloud in a female Hindi voice, using the phone's own voice or Amazon Polly's Kajal.
Voice notes	Speak in Hindi on WhatsApp; JalTaap understands (Amazon Transcribe) and replies by voice too.
No app needed	Works inside WhatsApp. No download, no sign-up. Share the answer on WhatsApp in one tap.

For site supervisors

Feature	What it does
Crew heat plan	For a crew of any size: which hours allow heavy work, rest breaks staggered in groups so work never stops, total litres of water to carry, ORS on very hot days. Printable.

For cities and agencies

Feature	What it does
AI-checked photo reports	An AI vision model checks each photo; only confident results go on the map.
Right office, automatically	Reports go to PWD, MCD, Delhi Jal Board, BMC, GHMC and others based on location and problem type.
Water-access reports	Tanker no-shows and dry taps count as confirmed when 3 people in the same area report them.
Officer view	Secure login; each office sees only its own reports, with photo, map and an AI action list.
Clear permission rules	Only the owning office can close a report, only verified reports can be closed, and disputed spots can be handed over (Cedar policies).
Control rooms	City control rooms can see every report in their city, read-only.

For everyone (the public website)

Feature	What it does
Live map	Heat, rain and flood spots for the next 48 hours across 63 cities, plus reports, hospitals and water points.
City dashboard	Heat now, safe hours, air, sun, rain and a 48-hour chart for any city.
Replays	Watch real past events: the 22 May 2026 heat and the 9 July 2026 storm in Delhi.
Hindi and English	The whole site switches language in one tap.

Coverage across India

Feature	What it does
63 cities	Delhi in full detail; every other million-plus city and every state and UT capital on the live map.
Anywhere in India	Any place name or location pin gets a personal heat plan.
Researched local data	Waterlogging spots for 10 big cities and heat-relief places for 17 cities, each with its news or government source; hospitals and drinking water from OpenStreetMap.

4. What makes JalTaap different

Many tools cover one piece: weather apps, government SMS alerts, city heat action plans, air-quality apps, complaint helplines. As far as we know, none of them combines these:

JalTaap	Most existing tools
Counts humid heat (wet-bulb), hour by hour	Heatwave declared on temperature only
A plan for your exact spot and your body	One warning for the whole district or city
Tells you WHEN you can work and how much to drink	Tells you how hot it is
Built for people who cannot read: colour, picture, voice	Text-heavy, mostly English
Works inside WhatsApp, no app	A separate app to download
Crew plan for supervisors	Nothing for employers
Heat, rain, air and water in one place	A different app for each
Report goes to the office that owns the spot, with an AI-checked photo	Complaint goes to a general helpline
Every number shows its data source and time; AI never decides the risk	Black-box alerts
63 cities and any point in India from day one	Usually one city or one state

5. Why JalTaap can win, judged on each criterion

What judges look at	How JalTaap answers it
Idea and impact: does it fix a real problem?	One clear user (outdoor workers), one clear gap (warnings that are not personal, local or actionable), proven with our own numbers: 414 humid-danger city-days a 40 °C rule missed.
Built on AWS	Deep, not decorative. AWS open source: Strands Agents (the AI agent), AWS SAM (the whole backend as code), Powertools, Cedar (permissions). AWS services: Bedrock (AI chat, photo checks), Lambda, API Gateway, SQS, EventBridge Scheduler, DynamoDB, S3, Amplify, Cognito, CloudWatch, Location Service, Transcribe, Polly, SES, Parameter Store, Budgets.
Design and usability: would an outsider know what to do?	A worker who cannot read gets one colour, one picture and a spoken answer. Judges get a live website, a map, a chat and replays, no login needed.
Execution: does it work?	196 automated tests pass. 34 test conversations check the AI's answers. The risk maths is tested against published values. Live forecast data flows through every screen.
Demo video	Strong scenes: a Hindi WhatsApp question and spoken reply, the 22 May heat replay, the 9 July storm replay, a photo becoming an officer task, and the AWS console.

6. Why you should select JalTaap

- **It protects lives.** Heat and floods kill Indian workers every year. JalTaap gives a clear, early, personal answer to the people at most risk.
- **It reaches people other tools leave out.** Voice, Hindi, pictures and WhatsApp mean even a worker who cannot read can use it.
- **It is honest and safe.** The AI never invents risk; a tested calculator does. Every answer shows its data time. Our limits are written down openly.
- **It closes the loop.** Reports do not vanish: they are checked, mapped and sent to the office that owns the spot, with clear rules about who can close them.
- **It is ready to grow.** 63 cities today; adding a city means adding a row of data, not new code. It runs on low-cost serverless AWS, so it can scale cheaply.
- **It is useful all year.** Heat in summer, flooding in the monsoon, polluted air in winter, water shortages all year.
- **It is proven, not just promised.** Our own backtests show the gap it fills, with the data and code published for anyone to check.

Honest limits

- Weather and air values are modelled forecasts, not station readings. The AQI is an estimate.
- Flood spots outside Delhi come mostly from news reports, and every map pin still needs a human check.
- JalTaap is not an official warning service; it adds to IMD and government advice, it does not replace it.

What comes next

- Official IMD warnings (the connection is built and waits for API access).
- Official hotspot lists from every city, and more languages: Bengali, Tamil, Marathi, Telugu.
- SMS for workers without WhatsApp, and partnerships with builders, delivery companies and city control rooms.

Built by Team DevCore for Environmental Hacks (WeMakeDevs × AWS Bharat Builds Tour), Heat & Water track, 8 to 11 October 2026. Weather data by Open-Meteo.com (CC BY 4.0). Map data © OpenStreetMap contributors.